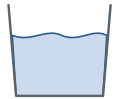
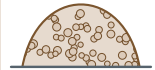
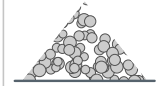
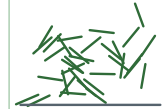
1 Primary shaping *(Urformen)*: from a shapeless material to a solid

	1.1 liquid *flüssig*	1.2 plastic *plastisch*	1.3 pasty *breiig*	1.4 powder *körnig oder pulverförmig*	1.5 fibres *span- oder faserförmig*	1.6 gas *gas- oder dampfförmig*	1.7 ionised *ionisiert*
state of the starting material							
becomes solid by	cooling and solidification; curing of resins	viscous flow, then cooling or vulcanisation	drying, hydration, gelation; firing (ceramics)	sintering (diffusion), binder, or melting	binder cured under pressure; fibre bonding	condensation or reaction on a hot surface	reduction of ions on a cathode
conventional	sand casting die casting investment casting continuous casting resin casting	injection moulding extrusion blow moulding thixomoulding (semi-solid metal)	slip casting clay extrusion concrete casting tape casting	pressing + sintering metal injection moulding HIP of powder sand moulds	particle board, MDF pressing papermaking felting	CVD, PVD free-standing CVD parts (SiC, diamond, Ni shells)	electroforming: Ni mould shells, foils, waveguides, micro-parts
additive	vat photo-polymerisation (SLA, DLP, 2PP) material jetting	material extrusion: filament (FFF), pellets, large-area	paste extrusion: robocasting, clay, concrete; ceramic slurry photopoly-merisation	powder bed fusion (laser, electron beam, SLS) binder jetting powder DED cold spray	sheet lamination (paper LOM) continuous-fibre extrusion	laser CVD beam-induced deposition aerosol jet	electrochemical AM: localised electro-deposition through a micro-nozzle
standard	*ISO/ASTM: VPP, MJT*	*MEX*	*MEX, VPP*	*PBF, BJT, DED*	*SHL, MEX*	*(no category)*	*(no category)*
at home	*ice cube, jelly*	*hot-glue gun*	*icing, cake batter*	*sandcastle, sugar cube*	*papier-mâché*	*frost on a window*	*electroformed leaf*